

Question 5

September 9, 2026

1 Part a

```
[1]: """Classical Hodgkin-Huxley model under constant current injection.
```

Python port of hh_sim.m. Same parameters, same shifted potential convention of the 1952 papers, so rest is near $V = 0$ and the spike peak near $V = 100$.

Call it and capture what you need:

```
    t, V, spk, rate = hh_sim()
```

Running the file directly (python hh_sim.py) prints the spike count and rate at the default $I_d = 7$. There is no MATLAB-style ans here, but t is 30001 elements of 0 to 300 in steps of 0.01, so avoid echoing the whole return value in a notebook cell.

Regression checks with default parameters:

$I_d = 7$ gives sustained firing at 58.3 Hz

$I_d = 5$ gives no sustained firing

The 58.3 is solver independent: BDF, LSODA, Radau and RK45 all agree to three digits, so a port that lands more than a Hz away has a real bug, most likely a sign in an exponent. The removable singularities in α_m and α_n at $V = 25$ and $V = 10$ are patched below; if you rewrite them, check that $\alpha_m(25)$ and $\alpha_n(10)$ come out finite.

The reported rheobase depends on how you define repetitive firing: both skipfrac and the spike count you demand of it will move the answer, because the underlying bifurcation is subcritical. State the criterion you used rather than treating the number as absolute.

```
"""
```

```
import numpy as np
```

```
from scipy.integrate import solve_ivp
```

```
__all__ = ["hh_sim"]
```

```
# ---- gating rate functions (ms-1), with removable singularities patched ----
```

```
def _am(V):
```

```

    return 1.0 if abs(V - 25) < 1e-6 else 0.1 * (25 - V) / (np.exp((25 - V) / 10) - 1)

def _bm(V):
    return 4 * np.exp(-V / 18)

def _ah(V):
    return 0.07 * np.exp(-V / 20)

def _bh(V):
    return 1 / (np.exp((30 - V) / 10) + 1)

def _an(V):
    return 0.1 if abs(V - 10) < 1e-6 else 0.01 * (10 - V) / (np.exp((10 - V) / 10) - 1)

def _bn(V):
    return 0.125 * np.exp(-V / 80)

def hh_sim(Id = 7.0, tmax = 300.0, gK = 36.0, gNa = 120.0, skipfrac = 0.4):
    """Integrate the HH model under constant current.
    Parameters
    -----
    Id : injected current density (uA/cm2). Defaults to 7, above
        rheobase, so hh_sim() with no arguments gives repetitive
        firing.
    tmax : simulation duration (ms).
    gK, gNa : maximal conductances (mS/cm2).
    skipfrac : fraction of the trace discarded before counting spikes, so
        that the onset transient is not mistaken for repetitive firing.

    Returns
    -----
    t, V : time vector (ms) and voltage trace (mV).
    spk : spike times, upward crossings of 40 mV.
    rate : mean firing rate in Hz over the counted window, 0 if fewer
        than two spikes.
    """
    C, gL, ENa, EK, EL = 1.0, 0.3, 115.0, -12.0, 10.6

    def rhs(_, y):
        V, m, h, n = y
        INa = gNa * m**3 * h * (V - ENa)
        IK = gK * n**4 * (V - EK)
        IL = gL * (V - EL)
        return [
            (Id - INa - IK - IL) / C,

```

```

        _am(V) * (1 - m) - _bm(V) * m,
        _ah(V) * (1 - h) - _bh(V) * h,
        _an(V) * (1 - n) - _bn(V) * n,
    ]

    V0 = 0.0 # steady state gating at rest
    y0 = [
        V0,
        _am(V0) / (_am(V0) + _bm(V0)),
        _ah(V0) / (_ah(V0) + _bh(V0)),
        _an(V0) / (_an(V0) + _bn(V0)),
    ]

    t = np.arange(0, tmax + 1e-9, 0.01)
    sol = solve_ivp(rhs, [0, tmax], y0, method = "BDF", t_eval = t,
                    rtol = 1e-8, atol = 1e-10, max_step = 0.05)
    V = sol.y[0]

    i0 = max(1, int(round(skipfrac * len(t))))
    up = np.flatnonzero((V[i0:-1] < 40) & (V[i0 + 1:] >= 40)) + i0
    spk = t[up]

    rate = 1000 * (len(spk) - 1) / (spk[-1] - spk[0]) if len(spk) >= 2 else 0.0
    return t, V, spk, rate

```

```

[2]: def fires(Id, gK = 36.0, nmin = 2):
    """Criterion: >= nmin spikes in the post-skip window of hh_sim."""
    _, _, spk, _ = hh_sim(Id = Id, gK = gK)
    return len(spk) >= nmin

def bisect(lo, hi, gK = 36.0, tol = 5e-4):
    while hi - lo > tol:
        mid = (lo + hi) / 2
        if fires(mid, gK = gK): hi = mid
        else: lo = mid
    return mid

rheobase = bisect(5.0, 7.0)
print(f"rheobase (300ms, nmin=2) = {rheobase:.3f}")

```

rheobase (300ms, nmin=2) = 6.256

```

[3]: import matplotlib.pyplot as plt
eps = 0.05
fig, axs = plt.subplots(2, 1, figsize = (7.5, 6), sharex = True, sharey = True)
for ax, dI in zip(axs, (-eps, +eps)):
    t, V, spk, rate = hh_sim(Id = rheobase + dI)
    ax.plot(t, V, "k", lw = 0.8)

```

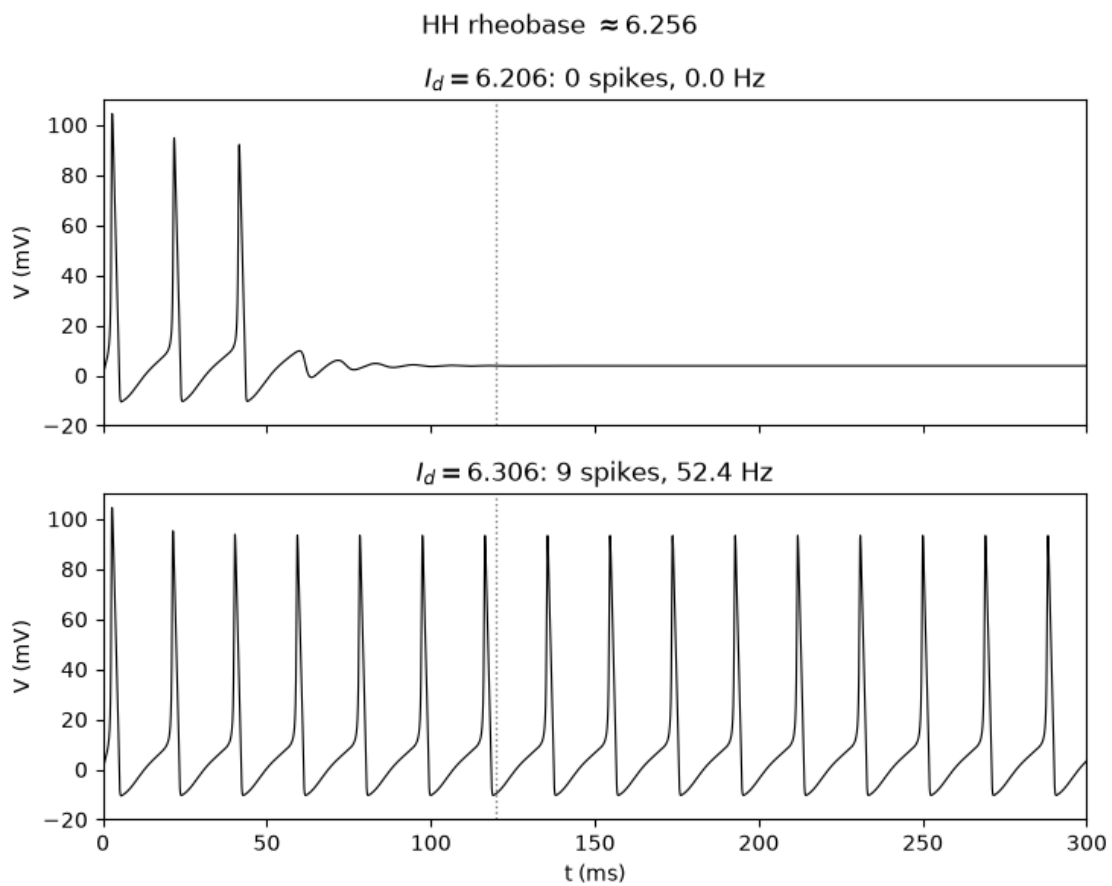
```

ax.axvline(120, color = "gray", ls = ":", lw = 1)
ax.set_title(f"$I_d = {rheobase + dI:.3f}$: {len(spik)} spikes, {rate:.1f} Hz")
ax.set_ylabel("V (mV)")

axs[0].set_ylim(-20, 110)
axs[1].set_xlim(0, 300)
axs[1].set_xlabel("t (ms)")
fig.suptitle(f"HH rheobase $\backslash\approx {rheobase:.3f}$")

fig.tight_layout()
plt.show()

```



2 Part (b)

```
[4]: Is = np.linspace(rheobase + 0.05, 2 * rheobase, 20)
rates = []
for Id in Is:
    t, V, spk, rate = hh_sim(Id = Id)
    rates.append(rate)
    print(f"I = {Id:6.3f}    spikes = {len(spk):3d}    rate = {rate:7.3f} Hz")

rates = np.array(rates)
```

```
I = 6.306    spikes = 9    rate = 52.432 Hz
I = 6.633    spikes = 10   rate = 56.099 Hz
I = 6.960    spikes = 11   rate = 58.096 Hz
I = 7.286    spikes = 10   rate = 59.674 Hz
I = 7.613    spikes = 11   rate = 61.024 Hz
I = 7.940    spikes = 11   rate = 62.243 Hz
I = 8.266    spikes = 11   rate = 63.359 Hz
I = 8.593    spikes = 12   rate = 64.403 Hz
I = 8.920    spikes = 12   rate = 65.383 Hz
I = 9.246    spikes = 12   rate = 66.317 Hz
I = 9.573    spikes = 13   rate = 67.204 Hz
I = 9.899    spikes = 12   rate = 68.057 Hz
I = 10.226   spikes = 12   rate = 68.879 Hz
I = 10.553   spikes = 12   rate = 69.669 Hz
I = 10.879   spikes = 12   rate = 70.441 Hz
I = 11.206   spikes = 13   rate = 71.183 Hz
I = 11.533   spikes = 13   rate = 71.908 Hz
I = 11.859   spikes = 13   rate = 72.613 Hz
I = 12.186   spikes = 13   rate = 73.300 Hz
I = 12.513   spikes = 14   rate = 73.973 Hz
```

```
[5]: fig, ax = plt.subplots(figsize = (6.5, 4.5))
ax.plot(Is, rates, "o-", lw = 1.2)
ax.annotate(f"onset $\approx$ {rates[0]:.3f}$ Hz", xy = (rheobase, rates[0]),
    xytext = (rheobase + 0.6, 10), arrowprops = dict(arrowstyle = "->"))
ax.set_xlabel("$I_d$ $\left(\frac{\mu A}{\text{cm}^2}\right)$")
ax.set_ylabel("firing rate $f$ (Hz)")
ax.set_xlim(min(Is) - 0.4, max(Is) + 0.4)
ax.set_ylim(0, rates.max() + 8)

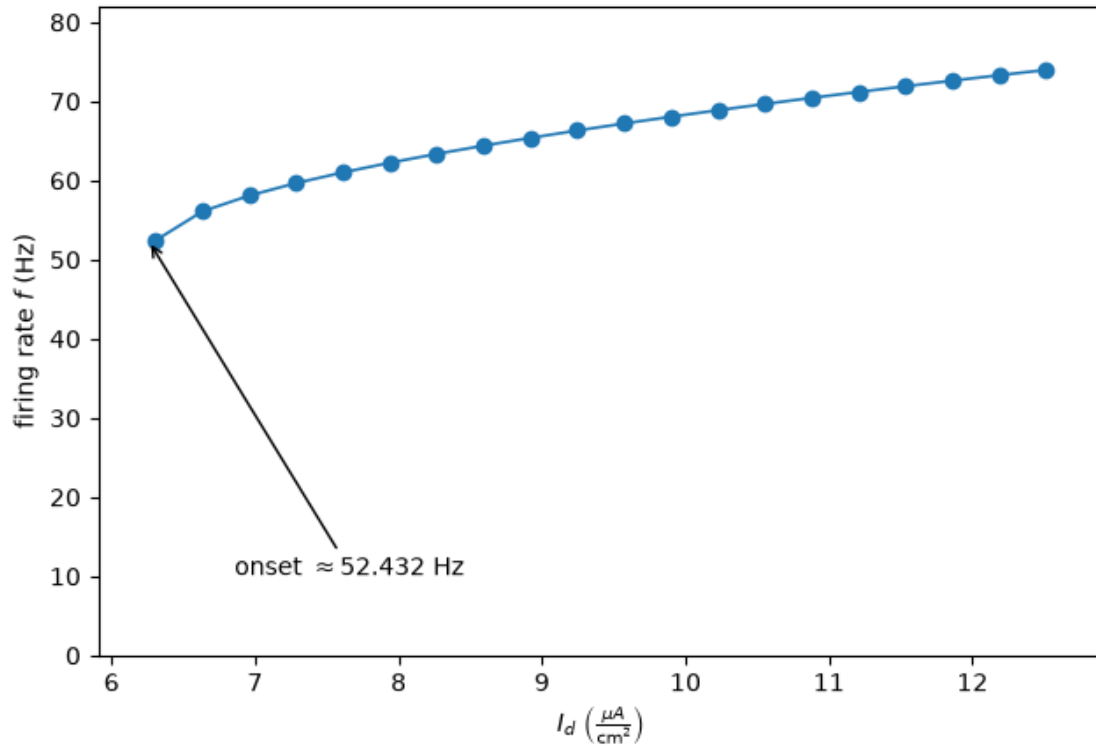
fig.tight_layout()
plt.show()
```

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<>:4: SyntaxWarning: invalid escape sequence '\m'
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<>:4: SyntaxWarning: invalid escape sequence '\m'
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```
ax.set_xlabel("$I_d$ $\left(\frac{\mu A}{cm^2}\right)$")
```



3 Part (c)

```
[6]: rheobase_lower_gK = bisect(5.0, 7.0, gK = 30.0)
print(f"rheobase (300ms, nmin=2, gK=30.0) = {rheobase_lower_gK:.3f}")
```

```
rheobase (300ms, nmin=2, gK=30.0) = 5.000
```

```
[7]: rates_lower_gK = []
for Id in Is:
    t, V, spk, rate = hh_sim(Id = Id, gK = 30.0)
    rates_lower_gK.append(rate)
    print(f"I = {Id:6.3f}    spikes = {len(spk):3d}    rate = {rate:7.3f} Hz")

rates_lower_gK = np.array(rates_lower_gK)
```

```
I = 6.306    spikes = 12    rate = 65.332 Hz
I = 6.633    spikes = 12    rate = 66.317 Hz
I = 6.960    spikes = 13    rate = 67.268 Hz
I = 7.286    spikes = 12    rate = 68.187 Hz
I = 7.613    spikes = 12    rate = 69.074 Hz
I = 7.940    spikes = 12    rate = 69.930 Hz
```

I = 8.266	spikes = 13	rate = 70.767 Hz
I = 8.593	spikes = 13	rate = 71.578 Hz
I = 8.920	spikes = 13	rate = 72.368 Hz
I = 9.246	spikes = 13	rate = 73.140 Hz
I = 9.573	spikes = 14	rate = 73.893 Hz
I = 9.899	spikes = 14	rate = 74.627 Hz
I = 10.226	spikes = 14	rate = 75.345 Hz
I = 10.553	spikes = 14	rate = 76.046 Hz
I = 10.879	spikes = 13	rate = 76.736 Hz
I = 11.206	spikes = 14	rate = 77.413 Hz
I = 11.533	spikes = 14	rate = 78.078 Hz
I = 11.859	spikes = 14	rate = 78.726 Hz
I = 12.186	spikes = 14	rate = 79.365 Hz
I = 12.513	spikes = 14	rate = 80.000 Hz

```
[8]: fig, ax = plt.subplots(figsize = (6.5, 4.5))
ax.plot(Is, rates, "o-", lw = 1.2, label = "gK = 36.0")
ax.plot(Is, rates_lower_gK, "o-", lw = 1.2, label = "gK = 30.0")

ax.legend()
ax.set_xlabel("$I_d$ $\left(\frac{\mu A}{\text{cm}^2}\right)$")
ax.set_ylabel("firing rate $f$ (Hz)")
ax.set_xlim(min(Is) - 0.4, max(Is) + 0.4)

fig.tight_layout()
plt.show()
```

```
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<>:6: SyntaxWarning: invalid escape sequence '\m'
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'\m'
ax.set_xlabel("$I_d$ $\left(\frac{\mu A}{\text{cm}^2}\right)$")
```

